Baseline Results

Reliable Hybrid SLC/MLC ReRAM Crossbars for Transformers

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1 What the baseline is

The baseline is **HyFlexPIM** (Song *et al.*, ISCA 2025), reproduced on our cell-level crossbar simulator. Every encoder linear layer of BERT-Base (72 matrices) and DistilBERT (36 matrices) is factorised with SVD, keeping $k = mn/(m+n)$ components so that storage is unchanged, and fine-tuned on SST-2 for 8 epochs. Components are ranked by the HyFlexPIM gradient score, $\sum_b |\partial L_b/\partial \sigma_r|$ over batches of 32 training sentences, and the top 20% of each layer go to SLC cells, the rest to MLC cells. Three reference chips are evaluated alongside it:

- **All SLC**: every component in reliable 1-bit cells (14 cells per weight), the most robust and most expensive chip;
- **All MLC**: every component in dense 2-bit cells (8 cells per weight), the cheapest chip;
- **Random 20%**: a random 20% of components in SLC, a control for whether the choice matters.

The hybrid chips use 9.2 cells per weight, 34% fewer than all-SLC. Accuracy is measured on all 872 SST-2 validation sentences and averaged over several simulated chips (random draws of all cell errors). Every placement is tested on the same chips. Errors are modelled per cell: 8-bit weights with one scale per output column, differential cell pairs, 7 SLC or 4 MLC bit slices per sign, and programming noise $\tilde{G} = G + \sigma G_{\max}\varepsilon$ on every cell.

2 Accuracy before hardware errors

Table 1: SST-2 accuracy (%) without hardware errors.

Model	FP32	INT8	SVD, truncated	SVD, fine-tuned
BERT-Base	92.43	91.86	89.33	90.14
DistilBERT	91.06	90.37	89.68	89.11

The SVD factorisation costs about two points after fine-tuning; this cost is common to every placement. On an error-free simulated chip every placement stays within about one point of the digital SVD model, confirming the simulator.

3 Programming noise

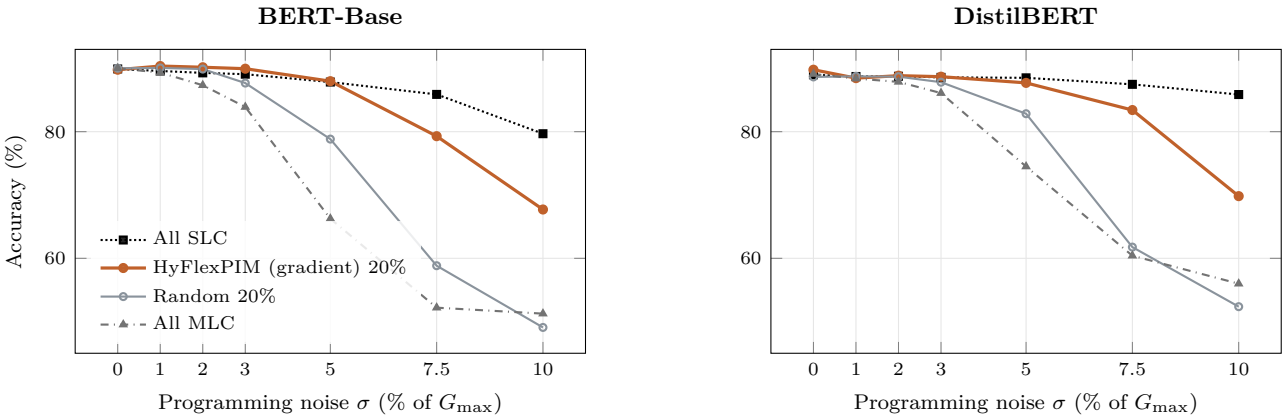


Figure 1: Accuracy against programming noise, mean of 5 chips.

Source: `02_programming_noise/results/noise/`.

Table 2: Accuracy (%) against programming noise σ, mean of 5 chips.

Model	Placement	0%	1%	2%	3%	5%	7.5%	10%
BERT-Base	All SLC	89.9	89.6	89.3	89.1	87.8	85.9	79.7
	All MLC	90.0	89.3	87.3	83.9	66.3	52.2	51.3
	Random 20%	90.0	90.1	89.9	87.7	78.8	58.8	49.1
	HyFlexPIM gradient 20%	89.8	90.4	90.2	90.0	88.0	79.3	67.7
DistilBERT	All SLC	89.0	88.7	88.8	88.6	88.5	87.5	85.9
	All MLC	89.1	88.4	87.9	86.1	74.5	60.4	56.0
	Random 20%	88.6	88.7	88.7	87.8	82.8	61.8	52.4
	HyFlexPIM gradient 20%	89.8	88.5	88.9	88.7	87.7	83.4	69.8

- **All-MLC fails at moderate noise**: BERT-Base falls to 66.3% at $\sigma = 5\%$ and to near chance (52.2%) at 7.5%. A hybrid layout is necessary.
- **The HyFlexPIM hybrid works at low noise but degrades at high noise**: with 20% SLC it matches all-SLC up to $\sigma = 5\%$, then trails it by 6.6 points at 7.5% and 12 points at 10% on BERT-Base.
- **Random selection recovers little**: which components are protected matters, not only how many.

4 SLC share

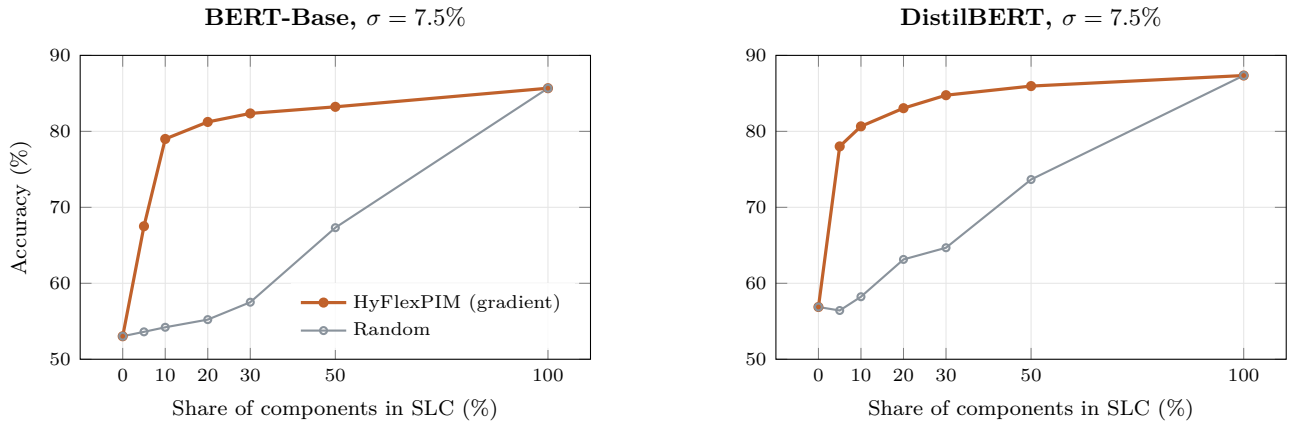


Figure 2: Accuracy against the share of components in SLC at $\sigma = 7.5\%$, mean of 20 chips; 0% is all-MLC and 100% all-SLC.

Source: `03_slc_share/results/slc_share/`.

Table 3: Accuracy (%) of the HyFlexPIM gradient placement and a random choice by share of components in SLC, mean of 20 chips.

Model	σ	Placement	0% (all MLC)	5%	10%	20%	30%	50%	100% (all SLC)
BERT-Base	3%	HyFlexPIM gradient	82.1	88.7	90.0	90.0	89.9	89.6	89.2
		Random	82.1	87.0	86.9	87.6	87.3	89.0	89.2
	5%	HyFlexPIM gradient	66.8	84.2	88.2	88.4	88.4	88.1	88.0
		Random	66.8	72.5	73.6	76.2	77.8	83.4	88.0
	7.5%	HyFlexPIM gradient	53.0	67.5	79.0	81.2	82.4	83.2	85.7
		Random	53.0	53.6	54.2	55.2	57.5	67.3	85.7
DistilBERT	3%	HyFlexPIM gradient	86.0	88.4	88.5	88.6	88.8	88.7	88.7
		Random	86.0	86.4	86.8	87.5	88.1	88.4	88.7
	5%	HyFlexPIM gradient	74.8	87.3	87.3	87.7	88.1	87.8	88.2
		Random	74.8	75.4	77.4	81.9	82.3	85.7	88.2
	7.5%	HyFlexPIM gradient	56.9	78.0	80.7	83.1	84.8	86.0	87.4
		Random	56.9	56.4	58.2	63.1	64.7	73.7	87.4

At $\sigma = 7.5\%$ the gradient placement does not reach all-SLC even with 50% of the components in SLC (83.2% against 85.7% on BERT-Base), so most of the cell savings of the hybrid are lost at high noise; a random choice needs about half of the components in SLC to recover at all.

5 Stuck cells

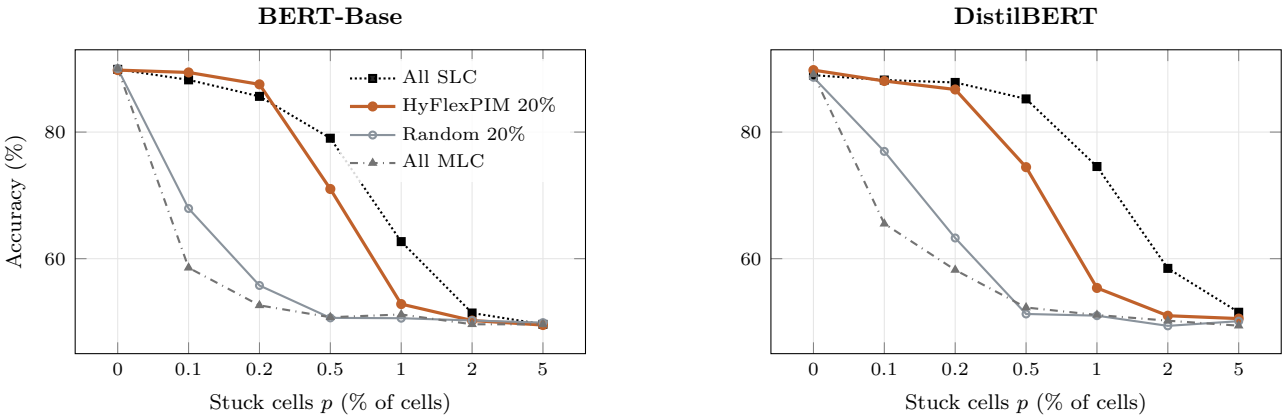


Figure 3: Accuracy against the share of stuck cells (half at $G_{\max}$, half at $G_{\min}$), no programming noise, mean of 20 chips. The horizontal axis is not to scale.

Source: `04_stuck_cells/results/fault/`.

Table 4: Accuracy (%) against the share of stuck cells p, no programming noise, mean of 20 chips.

Model	Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
BERT-Base	All SLC	89.9	88.3	85.6	79.0	62.7	51.4	49.6
	All MLC	90.0	58.6	52.6	50.8	51.2	49.6	49.6
	Random 20%	90.0	67.9	55.8	50.7	50.6	50.2	49.9
	HyFlexPIM gradient 20%	89.8	89.4	87.5	71.0	52.8	50.2	49.5
DistilBERT	All SLC	89.0	88.2	87.8	85.2	74.6	58.5	51.5
	All MLC	89.1	65.5	58.2	52.3	51.1	50.2	49.4
	Random 20%	88.6	76.9	63.3	51.3	51.0	49.4	50.1
	HyFlexPIM gradient 20%	89.8	88.1	86.7	74.5	55.4	51.0	50.5

- **Stuck cells break every baseline**: all-MLC fails at 0.1% stuck cells (58.6% on BERT-Base), the HyFlexPIM hybrid drops to 71.0% at 0.5% and 52.8% at 1%, and even all-SLC falls to 62.7% at 1%.
- A stuck cell in a high-order slice shifts its weight by a large fixed amount, so faults are far more harmful than noise, and accuracy varies widely between chips depending on where the faults fall.

6 Other non-idealities

Each effect is applied on its own, with the others disabled. Source: `05_other_non_idealities/results/<experiment>/`.

Table 5: Read noise σ_r (5 chips).

Model	Placement	0%	0.5%	1%	2%	3%	5%
BERT-Base	All SLC	89.7	89.8	89.8	89.0	87.2	80.0
	All MLC	90.4	89.4	87.7	77.3	63.8	53.0
	HyFlexPIM gradient 20%	90.3	90.5	90.3	89.8	88.3	76.6
DistilBERT	All SLC	88.5	89.0	88.2	88.3	87.5	84.5
	All MLC	89.0	88.6	87.9	83.8	74.5	57.7
	HyFlexPIM gradient 20%	88.6	88.9	88.5	88.1	86.8	80.5

Table 6: MLC level-dependent noise at $\sigma = 5\%$ (10 chips).

Model	Placement	uniform	mid ×1.5	mid ×2
BERT-Base	All SLC	88.2	88.2	88.2
	All MLC	67.3	64.7	62.5
	HyFlexPIM gradient 20%	88.3	88.1	87.8
DistilBERT	All SLC	88.4	88.4	88.4
	All MLC	75.7	73.5	70.3
	HyFlexPIM gradient 20%	88.0	87.5	87.1

Table 7: Fault patterns at 1% stuck cells (10 chips).

Model	Placement	scattered	clusters	lines
BERT-Base	All SLC	61.3	66.7	57.0
	All MLC	50.7	50.8	50.4
	HyFlexPIM gradient 20%	52.6	51.5	51.0
DistilBERT	All SLC	74.2	80.3	71.2
	All MLC	51.2	51.2	50.8
	HyFlexPIM gradient 20%	55.0	56.4	52.2

Table 8: IR drop, wire resistance r (deterministic).

Model	Placement	0	10^{-4}	3×10^{-4}	10^{-3}	3×10^{-3}
BERT-Base	All SLC	89.7	90.3	88.9	87.0	58.5
	All MLC	90.4	89.3	89.8	90.6	79.1
	HyFlexPIM gradient 20%	90.3	90.5	90.5	88.6	76.4
DistilBERT	All SLC	88.5	89.9	89.4	85.3	70.1
	All MLC	89.0	88.6	89.1	89.9	77.9
	HyFlexPIM gradient 20%	88.6	89.6	89.6	86.9	74.7

Table 9: Fixed ADC with bit-serial inputs (deterministic).

Model	Placement	ideal	8 bit	7 bit	6 bit	5 bit	4 bit	3 bit
BERT-Base	All SLC	89.7	89.9	90.0	89.3	51.1	61.1	51.1
	All MLC	90.4	89.6	89.2	83.0	51.1	60.3	50.9
	HyFlexPIM gradient 20%	90.3	90.1	90.1	89.6	53.0	63.2	51.9
DistilBERT	All SLC	88.5	89.2	88.4	88.0	71.0	52.1	59.1
	All MLC	89.0	88.3	87.8	81.7	56.5	51.9	56.8
	HyFlexPIM gradient 20%	88.6	88.4	88.5	88.4	77.3	51.5	60.1

Table 10: Fixed ADC with $\sigma = 5\%$ programming noise (3 chips).

Model	Placement	ideal	8 bit	6 bit	4 bit
DistilBERT	All SLC	88.1	88.3	88.7	52.6
	All MLC	77.8	76.8	56.0	53.6
	HyFlexPIM gradient 20%	87.8	87.6	87.2	51.8

- Read noise and level-dependent noise behave like programming noise: all-MLC fails, and the HyFlexPIM hybrid loses 3.4–4 points against all-SLC at 5% read noise.
- A fixed 6-bit ADC is enough for all-SLC and the hybrid; all-MLC needs 7 bits; 5 bits fails BERT-Base.
- IR drop has no effect at moderate wire resistance and hurts SLC most at high resistance, because a stored SLC one sits at the highest conductance.
- Every fault pattern leaves the HyFlexPIM hybrid close to chance at 1% stuck cells.

7 Second dataset: MNLI

The same baselines on MNLI (three classes, chance 33.3%), on 2,000 matched validation pairs; the digital SVD models reach 83.2% (BERT-Base) and 80.9% (DistilBERT). On MNLI the simulator loses accuracy even on an

error-free chip (up to 11 points for all-SLC on BERT-Base) because the vector between a tier's two crossbars is re-quantised with a single 8-bit scale; only placements with the same tier sizes should therefore be compared directly. Source: `08_mnli/results/{slc_share,fault}/`.

Table 11: MNLI accuracy (%) by share of components in SLC at $\sigma = 5\%$, mean of 20 chips.

Model	σ	Placement	0% (all MLC)	5%	10%	20%	30%	50%	100% (all SLC)
BERT-Base	5%	HyFlexPIM gradient	33.5	59.6	63.5	63.3	66.5	64.2	61.6
		Random	33.5	34.1	33.9	35.9	38.9	44.9	61.6
DistilBERT	5%	HyFlexPIM gradient	36.8	66.6	69.0	71.3	72.1	73.6	72.0
		Random	36.8	38.3	39.3	44.1	48.1	55.3	72.0

Table 12: MNLI accuracy (%) against the share of stuck cells, 20% SLC, mean of 20 chips.

Model	Placement	0%	0.1%	0.2%	0.5%	1%	2%	5%
BERT-Base	All SLC	72.2	67.2	61.9	44.4	34.5	32.4	32.4
	All MLC	72.4	37.2	33.2	32.8	31.8	31.9	32.3
	Random 20%	73.0	42.8	34.3	33.3	32.6	32.4	33.5
	HyFlexPIM gradient 20%	78.2	73.0	65.2	36.9	32.0	31.5	32.4
DistilBERT	All SLC	76.9	74.4	71.3	60.4	42.1	34.1	34.2
	All MLC	76.8	44.5	34.0	33.6	33.7	34.1	33.0
	Random 20%	77.7	51.6	35.3	32.9	33.2	33.7	33.4
	HyFlexPIM gradient 20%	79.6	76.0	71.4	44.5	33.9	32.9	33.9

8 Summary of the baseline

- A hybrid SLC/MLC layout is necessary: all-MLC fails at moderate programming noise and at 0.1% stuck cells.
- The HyFlexPIM gradient placement works at low noise but falls behind all-SLC at $\sigma \geq 7.5\%$, by 4–16 points at 20% SLC and by up to 18 points at 5% SLC on BERT-Base.
- No baseline handles stuck cells: at 1% stuck cells the HyFlexPIM hybrid is near chance and all-SLC loses 14–27 points.

These gaps are what our contributions address; see the separate *Contribution Results* document.